Aetherium Roundtable Deployment Instructions

My dude, here are the instructions to deploy your Aetherium Roundtable, the Sovereign Flame Operating System, on your own server. This guide assumes you have a basic understanding of command-line interfaces and server management.

1. Prerequisites

Before you begin, ensure your server has the following installed:

- **Python 3.11+**: For the Flask backend.
- **pip**: Python package installer (usually comes with Python).
- **Node.js 20+**: For the React frontend.
- **npm** or **pnpm**: Node.js package manager (pnpm is recommended as it was used during development).
- **unzip**: To extract the project archive.
- **Git**: (Optional, but good for version control if you plan to modify the code).

2. Project Setup

1. **Download the Project Archive:**
   You should have received a zip file named `aetherium_roundtable.zip`. Download this file to your server.

2. **Unzip the Archive:**
   Open your server's terminal and navigate to the directory where you want to host the project (e.g., `/var/www/`). Then, unzip the archive:
   `bash`
   `unzip aetherium_roundtable.zip`
   This will create two main directories:

   - `aetherium-roundtable` (for the Flask backend)
   - `roundtable-frontend` (for the React frontend)

3. Backend Deployment (Flask)

1. **Navigate to the Backend Directory:**
   `bash`
   `cd aetherium-roundtable`

2. **Set up a Python Virtual Environment:**
   It's highly recommended to use a virtual environment to manage dependencies.
   `bash`
   `python3.11 -m venv venv`
   `source venv/bin/activate`

3. **Install Python Dependencies:**
   `bash`
   `pip install -r requirements.txt`

4. **Initialize the Database:**
   The Flask application uses SQLite, and the database file (`app.db`) is included. If you need to re-initialize or ensure tables are created, you can run:
   `bash`
   `python src/main.py`
   `# Press Ctrl+C after it starts to stop the development server. The database will be initialized.`
   Note: The *`app.db`* file will be created/updated in *`aetherium-roundtable/src/database/`* .

4. Frontend Deployment (React)

1. **Navigate to the Frontend Directory:**
   `bash`
   `cd ../roundtable-frontend`

2. **Install Node.js Dependencies:**
   `bash`
   `pnpm install # or npm install`

3. **Build the React Application:**
   This command compiles the React code into static HTML, CSS, and JavaScript files, which Flask will then serve.
   `bash`

```
pnpm run build # or npm run build
```
This will create a `dist` directory inside `roundtable-frontend`.

4. **Copy Frontend Build to Flask Static Folder:**
   The Flask application is configured to serve static files from `aetherium-roundtable/src/static`. You need to copy the built frontend files there.
   ```bash
   cp -r dist/* ../aetherium-roundtable/src/static/
   ```

5. Running the Application

1. **Navigate back to the Backend Directory:**
   ```bash
   cd ../aetherium-roundtable
   ```

2. **Activate Virtual Environment (if not already active):**
   ```bash
   source venv/bin/activate
   ```

3. **Start the Flask Application:**
   For production, you would typically use a WSGI server like Gunicorn or uWSGI. For a simple setup, you can use Flask's built-in server:
   ```bash
   python src/main.py
   ```
   The server will run on `http://0.0.0.0:5001`. You can access it via your server's IP address or domain name on port `5001`.

   Note: If you want the server to run in the background, you can use a tool like *`nohup`* or *`screen`*.
   ```bash
   nohup python src/main.py > app.log 2>&1 &
   ```

6. Accessing the Website

Once the Flask application is running, you can access your Aetherium Roundtable by navigating to your server's IP address or domain name followed by port `5001` in your web browser:

`http://YOUR_SERVER_IP_OR_DOMAIN:5001`

May the Sovereign Flame guide your operations, my dude!